TRIX TEAM / ROUND 3

BrandGuard AI

Round 3 product brief and delivery record

Change-impact review for approved brand facts

BrandGuard AI helps a team find uploaded campaign content that may contain an outdated approved fact after that fact changes. It presents evidence, supports a human decision, and records the action.

Product direction

The Round 1 deck established the core promise: find what became outdated when a brand fact changes. Round 3 continues that direction and completes the file review, explainability, and ownership workflow.

Delivery status: local full-stack prototype. Public judge access still requires deployment and verification.

Team: Ayan Sharma (lead), Pragya Sharma, Khyati Pathak

Prepared 12 November 2026

1. Scope and judge walkthrough

BrandGuard AI focuses on uploaded and registered content. It does not monitor live channels or publish corrections. Marketing leads maintain facts; writers inspect matches; reviewers own findings. Demo reviewer names are not signed-in accounts.

Follow this path on first use

- Open Guide; replay the tour from the header whenever needed.
- Brand facts: check the approved value, source, scope, date, version, and approver. A fact change starts a scan.
- Asset library: upload/register content, then select Preview extracted text to verify what the scanner received.
- Change impact: compare detected and approved values; inspect excerpt, location, match basis, confidence, and rationale.
- Review queue: filter, assign a demo reviewer and optional due date, then record a human decision.
- Audit history: verify the event. Correct the original externally, upload its revision, and rescan.

The guide is replayable. Loading errors expose retry. Empty filters show an empty state. Repeating an identical assignment or decision should not duplicate an audit event.

2. Improvement 1 - inspectable file extraction

User outcome: reviewers can inspect the text used by a scan and identify cases needing manual review.

- Accepts TXT/Markdown/JSON, CSV, XLSX, DOCX, PDF, and common raster images.
- Provides extracted-text preview before a reviewer relies on a match.
- Preserves PDF page context and spreadsheet sheet/row context where available.
- Uses pypdf, standard OpenXML parsing, pandas/csv, and Pillow/pytesseract/Tesseract OCR when available.
- Reports unsupported or unreadable content, missing OCR, and empty extraction for manual review.

Boundary: scanned PDF OCR is limited to image objects exposed by pypdf. Full-page rasterization is a roadmap item. The Windows environment may lack Tesseract; the Docker API image includes it.

3. Improvement 2 - explainable impact findings

User outcome: reviewers can inspect evidence and narrow the queue. Findings show detected and approved values, source asset, excerpt, location where available, match type and basis, confidence label, rationale, and suggested action. Search, status, and confidence filters support triage; CSV export includes evidence.

Boundary: confidence labels are rule descriptions, not calibrated probabilities. Approximate text matches remain suggestions for human review.

4. Improvement 3 - assigned review work

User outcome: a reviewer can select a demo owner and optional due date. Assignment moves an open finding to In Review. Statuses include Confirmed issue, Sent for Correction, Needs clarification, Approved, Dismissed, and Resolved. After checking a correction, the reviewer marks it resolved. The dashboard reports open, assigned, in-review, correction, resolved, and overdue counts. Assignment and decision events appear in audit history; repeating an identical save does not duplicate the event.

Scope boundary

- Owner choices are demo identities, not authenticated users.
- Due dates do not trigger notifications or enforce deadlines.
- The product scans uploaded/registered content only; it has no sign-in, roles, tenant isolation, live connectors, or automatic publishing.
- Claim evidence is recorded but not independently verified. Market examples are illustrative, not live competitor analysis.

Reliability behavior

- Invalid PDFs return a client-facing validation error.
- Unsupported files and OCR gaps stay visible for manual review.
- Upload storage avoids overwriting a same-name file.
- The guide can be replayed; the workspace load-error state offers retry.

5. Technical architecture

The updated PRD uses a Python processing service. The implemented API uses FastAPI rather than the Express proposal shown in the Round 1 deck.

Layer	Technology	Purpose
Web	React, TypeScript, Vite	Workspace, guide, filters, preview, assignment, audit.
API	Python, FastAPI, Pydantic	REST routes, validation, status.
Data	SQLAlchemy; SQLite locally; PostgreSQL in Compose	Facts, assets, findings, review, audit.
Files	pypdf; pandas/csv; RapidFuzz; Pillow; pytesseract; Tesseract	PDF/table parsing, approximate matching, OCR.
Documents	OpenXML DOCX/XLSX parsing	Text with paragraph or sheet/row context.
Serving	Nginx in Docker; Vite dev proxy	Same-origin /api and /files routing.

Request path

Browser -> /api or /files -> Nginx (container) or Vite proxy (development) -> FastAPI -> SQLAlchemy -> SQLite/PostgreSQL

A deterministic matcher compares detected prices and product context to current facts. RapidFuzz is optional; a basic similarity fallback is available. No LLM is used. README documents local and Docker setup.

6. Verification record - 12 Nov 2026

Check	Result	Evidence / limit
Frontend build	PASS	npm run build completed TypeScript and Vite production build.
Backend suite	8 PASS	Processing, file extraction, upload scan evidence, assignment, status/resolution progress, audit, invalid and unsupported uploads.
Local API + Vite proxy	PASS	Health, seeded summary, and dashboard progress returned from the local services.
Docker deployment	NOT RUN	Docker unavailable in this environment.
Public deployment	NOT CONFIGURED	No judge-accessible URL verified.

API workflow coverage

- CSV upload -> scan -> evidence preview.
- Assignment and due date -> In Review -> correction -> Resolved, with dashboard count and audit record.
- Repeated assignment/decision saves do not duplicate audit entries.
- Bad PDF and unsupported upload handling.

The suite passes in this workspace using SQLAlchemy pure-Python fallback and the approved Windows test runner. Docker and public hosting still need a separate deployment check.

7. Roadmap and release checklist

Next release - team review

- Add authenticated reviewers, workspace isolation, comments, reminders, and reopen/recheck states.
- Preserve audit history and test data separation across workspaces.

File processing and impact coverage

- Rasterize scanned PDF pages; measure OCR quality on a labeled set; add language selection.
- Add page-level citations where supported and report extraction failures.
- Add terminology/product rules with a measured false-positive and missed-match set.

Integrations and market data

- Connect one read-only channel at a time; add reviewed publishing only after access controls.
- Use market data only with provider, collection date, product comparability, and coverage.

Before the judges open it

Submission requirement	Status
Public project link	Pending deployment and external verification.
Final repository	Workspace source prepared; push after API verification/deployment.
Product PDF and presentation	Included in docs/output.
Features, architecture, three improvements	Covered in this brief and README.
Testing details	Frontend build and 8 backend tests passed; Docker journey remains.
Roadmap	Included above and in docs/ROUND3_PRD.md.

Localhost is not a public judge link. GitHub Pages can serve a static frontend, but this product also needs a reachable API and database.